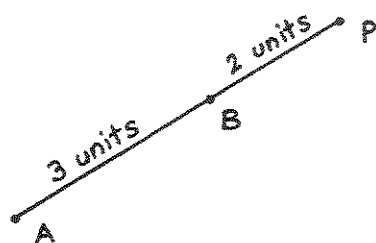


QUESTION 1

Marks

- a) Find $\frac{d}{dx} \left(\frac{1}{4+x^2} \right)$ 2
- b) Find $\int \frac{1}{4+x^2} dx$ 2
- c) Solve $\frac{1-x}{1+x} \leq 1$ 3
- d) The polynomial equation $3x^3 - 2x^2 + 3x - 4 = 0$ has roots α, β and δ . 3
- Find the value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\delta} + \frac{1}{\beta\delta}$

- e) 2

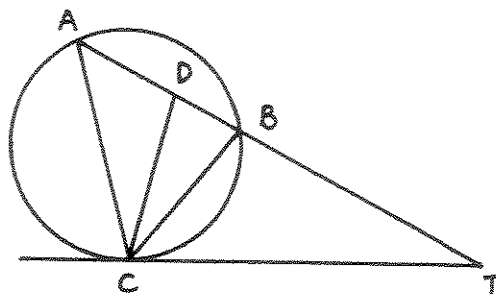


The point P divides the interval AB in the ratio 1:K. Find K

QUESTION 2

Marks

- a) Show that $\sin x - \cos 2x = 2 \sin^2 x + \sin x - 1$ 5
 Hence or otherwise solve
 $\sin x - \cos 2x = 0$ for $0 \leq x \leq 2\pi$
- b) ABC is a triangle inscribed in a circle. The tangent at C meets AB at T. 3
 The bisector of $\angle ACB$ cuts AB at D
- i) Copy the diagram
- ii) Prove $TC = TD$



Question 2 (cont.)

- c) Consider the sequence

4

$$\log_{10} (x-2), \quad \log_{10} (x-2)^2, \quad \log_{10} (x-2)^3$$

- i) Is this sequence arithmetic or geometric? Justify your answer.

- ii) Show that the sum to n terms is given by

$$\frac{n}{2} \log_{10} (x-2)^{n+1}$$

QUESTION 3

Marks

- a) i) Find x such that $\sin^{-1} x = \cos^{-1} x$

4

- ii) On the same number plane, sketch the graphs of

$$y = \sin^{-1} x \text{ and } y = \cos^{-1} x.$$

Label important points clearly.

- iii) On the same diagram as ii, sketch $y = \sin^{-1} x + \cos^{-1} x$

- b) i) By considering the graph of $y = e^x$ show that the equation

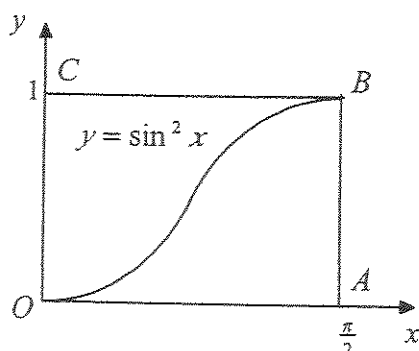
4

$$e^x + x + 1 = 0 \text{ has only one real root and that this root is negative.}$$

- ii) Taking $x = -1.5$ as a first approximation to this root, use one application of Newton's method to find a better approximation.

- c)

4



The rectangle $OABC$ has vertices $O(0,0)$, $A(\frac{\pi}{2}, 0)$, $B(\frac{\pi}{2}, 1)$ and $C(0,1)$.

The curve $y = \sin^2 x$ is shown passing through the points O and B . Show that this curve divides the rectangle $OABC$ into two regions of equal area.

QUESTION 4

Marks

- a) Prove by Mathematical Induction that

5

$$1 \times 3 + 2 \times 3^2 + \dots + n \times 3^n = \frac{(2n-1) 3^{n+1} + 3}{4}$$

where n is an interger, $n \geq 1$

- b) If $\tan^{-1} y = 2 \tan^{-1} x$ show that

3

$$y = \frac{2x}{1-x^2}$$

- c) Using the substitution $u = e^{2x}$ find a and b such that

4

$$\int_0^{\ln 2} \frac{e^{2x}}{1+e^{4x}} dx = \frac{1}{2} \tan^{-1} a - b$$

QUESTION 5

Marks

- a) The rate at which a body cools in air is proportional to the difference between its temperature T and the constant temperature 20°C (in this case) of the surrounding air. This can be expressed by the differential equation

6

$$\frac{dT}{dt} = -k (T - 20)$$

The original temperature of a heated metal bar was 100°C . The bar cools to 70°C in 10 minutes.

- Show that $T = 20 + Ae^{-kt}$ is a solution to the differential equation.
- Show $A = 80$
- Find the exact value of k
- Find the time taken for the temperature of the bar to reach 60°C .
(Give your answer to the nearest minute).

- b) $M(2am, am^2)$ and $N(2an, an^2)$ are points on the parabola $x^2 = 4ay$
- Find the equation of the chord MN
 - Find the co-ordinates of the midpoint of the chord MN
 - If the chords all pass through the point $(0,2)$, show that the locus of the midpoint of MN is

$$x^2 = 2a(y - 2)$$

6

QUESTION 6

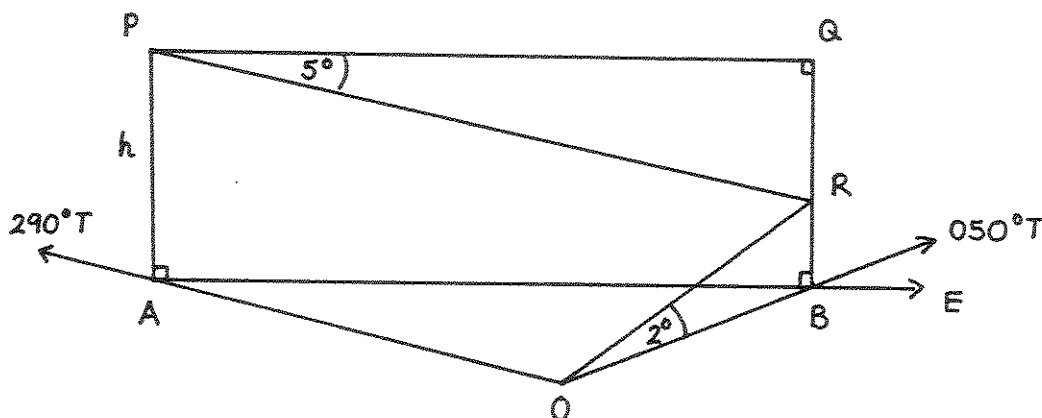
Marks

- a) $P(x)$ is an odd polynomial of degree 3. It has $x+4$ as a factor, and when it is divided by $x-3$ the remainder is 21. Find $P(x)$.

3

- b)

5



In the diagram above an aircraft is flying along the path PR. It has a constant speed of 300 km/h and is descending at a steady angle of 5° . It flies directly over beacons at A and B where B is due East of A. An observer at O first sights the aircraft over A at a bearing of 290°T . The observer sights the aircraft again 10 minutes later over B at a bearing of 050°T and with an angle of elevation of 2° . O is on the same horizontal plane as A and B.

- Show that the aircraft has travelled 50km in the 10 minutes between observations.
- Show that $\angle AOB = 120^\circ$.
- Prove that the observer at O is 19 670 metres, to the nearest 10 metres, from the beacon at B.
- Find the altitude h of the aircraft, to the nearest 10 metres, when it was originally sighted over A.

Question 6 (cont.)

c) If $f(x) = u(x) - \ln[u(x)+1]$

4

i) Show that $f'(x) = \frac{u(x)u'(x)}{1+u(x)}$

ii) Hence or otherwise evaluate

$$\int_0^{\frac{\pi}{2}} \frac{\cos x \sin x}{1 + \sin x} dx$$

QUESTION 7

Marks

a) The velocity v m/s of a particle at time t seconds is given in terms of position x m by

6

$$v = \frac{4}{x} \text{ (where } x > 0 \text{)}$$

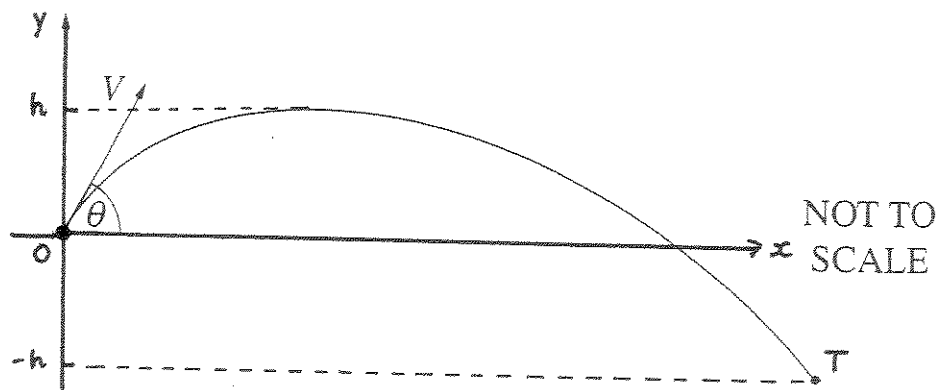
Initially $x = 8$.

i) Find the acceleration of the particle when $x = 1$

ii) Find an expression for x in terms of t .

iii) What is the position of the particle when $t = 2$?

iv) Describe the motion of the particle.



The diagram above shows the path of a projectile fired from the top O of a cliff. Its initial velocity is V m/s, its initial angle of elevation is θ and it rises to a maximum height h metres above O . It strikes a target T situated on a horizontal plane h metres below O .

The horizontal and vertical components of displacement in metres at time t seconds are given by $x = Vt \cos \theta$ and $y = Vt \sin \theta - \frac{1}{2}gt^2$ respectively.

i) Prove that $h = \frac{V^2 \sin^2 \theta}{2g}$.

ii) Prove that the time taken for the projectile to reach its target is:

$$\frac{V \sin \theta (1 + \sqrt{2})}{g} \text{ seconds}$$

iii) Hence show that the distance from the target to the base of the cliff is:

$$\frac{V^2 (1 + \sqrt{2}) \sin 2\theta}{2g} \text{ metres}$$

Question 1

$$a) \frac{d}{dx} \left(\frac{1}{4+x^2} \right)$$

$$= \frac{d}{dx} (4+x^2)^{-1}$$

$$= -1(4+x^2)^{-2} \cdot 2x$$

$$= \frac{-2x}{(4+x^2)^2}$$

$$b) \int \frac{1}{4+x^2} dx = \frac{1}{2} \tan^{-1} \frac{x}{2} + c$$

$$c) \frac{1-x}{1+x} \leq 1$$

$$(1+x)^2 \times \frac{1-x}{1+x} \leq 1 \times (1+x)^2$$

$$(1+x)(1-x) \leq (1+x)^2$$

$$(1+x)(1-x) - (1+x)^2 \leq 0$$

$$(1+x)[(1-x) - (1+x)] \leq 0$$

$$(1+x)(-2x) \leq 0$$



$$\therefore \underline{\underline{x < -1, x \geq 0}}$$

$$d) 3x^3 - 2x^2 + 3x - 4 = 0$$

$$\alpha + \beta + \delta = \frac{2}{3}$$

$$\alpha\beta\delta = \frac{4}{3}$$

$$\begin{aligned} \frac{1}{\alpha\beta} + \frac{1}{\alpha\delta} + \frac{1}{\beta\delta} &= \frac{\delta + \beta + \alpha}{\alpha\beta\delta} \\ &= \frac{\frac{2}{3}}{\frac{4}{3}} \\ &= \underline{\underline{\frac{1}{2}}} \end{aligned}$$

$$c) 5 : -2 = 1 : K$$

$$\therefore \underline{\underline{K = -\frac{2}{5}}}$$

Question 2

$$a) \text{LHS} = \sin x - \cos 2x$$

$$= \sin x - (1 - 2\sin^2 x)$$

$$= \sin x - 1 + 2\sin^2 x$$

$$= 2\sin^2 x + \sin x - 1$$

$$= \text{RHS}$$

$$\sin x - \cos 2x = 0$$

$$2\sin^2 x + \sin x - 1 = 0$$

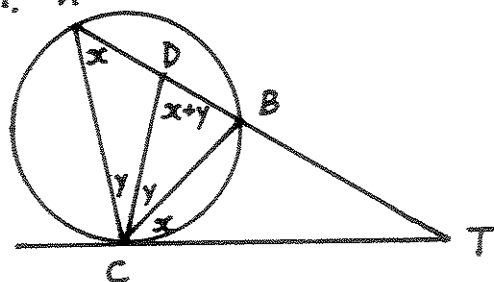
$$(2\sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2} \quad \text{or} \quad \sin x = -1$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6} \quad x = \frac{3\pi}{2}$$

$$\therefore \underline{\underline{x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}}}$$

b) i. A



ii. let $\angle BCT = x$

$$\angle BCD = \angle DCA = y$$

(given CD bisects $\angle ACB$)

$$\angle CAB = x \quad (\text{alternate segment theorem})$$

$$\angle TCD = x+y$$

$$\angle TDC = x+y \quad (\text{exterior angle of } \triangle ACD)$$

$\therefore TC = TD$ (equal sides are opposite equal angles)

c) $\log_{10}(x-2), 2\log_{10}(x-2), 3\log_{10}(x-2)$

i. arithmetic sequence

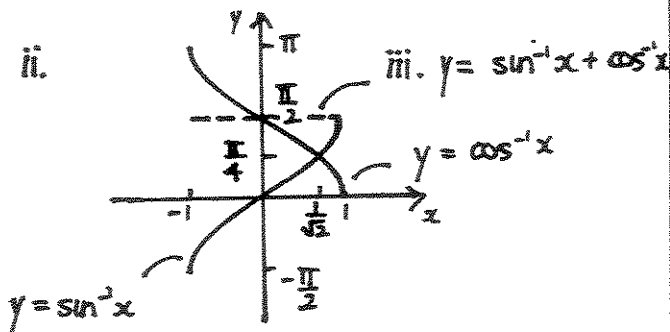
$$d = \log_{10}(x-2)$$

$$T_2 - T_1 = T_3 - T_2$$

$$\begin{aligned} \text{ii. } S_n &= \frac{n}{2} [2a + (n-1)d] \\ &= \frac{n}{2} [2\log_{10}(x-2) + (n-1)\log_{10}(x-2)] \\ &= \frac{n}{2} [\log_{10}(x-2)^2 + \log_{10}(x-2)^{n-1}] \\ &= \frac{n}{2} [\log_{10}(x-2)^{2+n-1}] \\ &= \frac{n}{2} \log_{10}(x-2)^{n+1} \end{aligned}$$

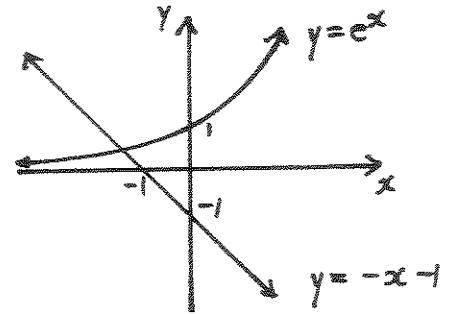
Question 3

a) i. $\underline{x = \frac{1}{\sqrt{2}}}$



iii. $y = \sin^{-1}x + \cos^{-1}x$
 $y = \pi$

b) i. Solution of $e^x + x + 1 = 0$ is the point of intersection of $y = e^x$ and $y = -x - 1$.



ie. one point of intersection with $x < -1$

$\therefore$ the equation has one real and negative root.

ii. see below

c) $A_{OABC} = \frac{\pi}{2} \times 1$
 $= \frac{\pi}{2}$

$$\begin{aligned} A_{OAB} &= \int_0^{\frac{\pi}{2}} \sin^2 x \, dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 - \cos 2x \, dx \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[\left(\frac{\pi}{2} - \frac{1}{2} \sin \pi \right) - \left(0 - \frac{1}{2} \sin 0 \right) \right] \\ &= \frac{1}{2} \times \frac{\pi}{2} \\ &= \frac{\pi}{4} \end{aligned}$$

$$\begin{aligned} A_{OBC} &= \frac{\pi}{2} - \frac{\pi}{4} \\ &= \frac{\pi}{4} \end{aligned}$$

$$\therefore A_{OBC} = A_{OAB}$$

$\therefore$ curve divides rectangle into two regions of equal area.

ii. $f(x) = e^x + x + 1$
 $f'(x) = e^x + 1$
 $x_2 = -1.5 - \frac{f(-1.5)}{f'(-1.5)}$
 $\therefore x_2 = -1.27$

Question 4

$$a) 1 \times 3 + 2 \times 3^2 + \dots + n \times 3^n \\ = \frac{(2n-1) 3^{n+1} + 3}{4}$$

Step 1: let $n=1$

$$\text{LHS} = 1 \times 3 \quad \text{RHS} = \frac{(2-1) 3^2 + 3}{4} \\ = 3 \quad = 3$$

$\therefore$ true for $n=1$

Step 2: assume true for $n=k$

$$\text{i.e. } S_k = \frac{(2k-1) 3^{k+1} + 3}{4}$$

Step 3: hence show true for $n=k+1$

i.e. show

$$S_{k+1} = \frac{[2(k+1)-1] 3^{k+1+1} + 3}{4} \\ = \frac{(2k+1) 3^{k+2} + 3}{4}$$

$$S_{k+1} = S_k + T_{k+1}$$

$$S_{k+1} = \frac{(2k-1) 3^{k+1} + 3}{4} + (k+1) 3^{k+1} \\ = \frac{(2k-1) 3^{k+1} + 3 + 4(k+1) 3^{k+1}}{4} \\ = \frac{3^{k+1} (2k-1 + 4k+4) + 3}{4} \\ = \frac{3^{k+1} (6k+3) + 3}{4} \\ = \frac{3^{k+1} \times 3^1 (2k+1) + 3}{4}$$

$$= \frac{3^{k+2} (2k+1) + 3}{4}$$

Step 4: Since true for $n=1$ then from step 3 true for $n=1+1=2$ and so on for all $n \geq 1$.

$$b) \tan^{-1} y = 2 \tan^{-1} x \\ \text{let } \alpha = \tan^{-1} y \quad \beta = \tan^{-1} x \\ \tan \alpha = y \quad \tan \beta = x$$

$$\text{So } \alpha = 2\beta$$

$$\tan \alpha = \tan 2\beta$$

$$\tan \alpha = \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

$$\therefore y = \frac{2x}{1-x^2}$$

$$c) u = e^{2x} \quad \text{when } x = \ln 2 \\ \frac{du}{dx} = 2e^{2x} \quad u = e^{2 \ln 2} \\ = 4 \\ \frac{du}{2} = e^{2x} dx \quad \text{when } x=0 \\ u = e^0 \\ = 1$$

$$\int_0^{\ln 2} \frac{e^{2x}}{1+e^{4x}} dx \\ = \frac{1}{2} \int_1^4 \frac{1}{1+u^2} du \\ = \frac{1}{2} [\tan^{-1} u]_1^4 \\ = \frac{1}{2} [\tan^{-1} 4 - \tan^{-1} 1] \\ = \frac{1}{2} \tan^{-1} 4 - \frac{\pi}{8} \\ \therefore a = 4 \quad \& \quad b = \frac{\pi}{8}$$

Question 5

a) i. $T = 20 + Ae^{-kt}$
 $\frac{dT}{dt} = -kAe^{-kt}$
 $\frac{dT}{dt} = -k(T-20)$ since $Ae^{-kt} = T-20$

ii. When $t=0$, $T=100$
 $100 = 20 + Ae^0$
 $A = 80$

iii. $T = 20 + 80e^{-kt}$
 When $t=10$, $T=70$
 $70 = 20 + 80e^{-10k}$
 $50 = 80e^{-10k}$
 $\frac{5}{8} = e^{-10k}$
 $\ln \frac{5}{8} = -10k$
 $\therefore k = \frac{1}{10} \ln \frac{5}{8}$

iv. When $T=60$, $t=?$
 $60 = 20 + 80e^{-kt}$
 $40 = 80e^{-kt}$
 $\frac{1}{2} = e^{-kt}$
 $t = \ln \frac{1}{2} \div -k$
 $\therefore t = 15 \text{ min}$

b) $M(2am, am^2)$
 $N(2an, an^2)$

i. $m_{MN} = \frac{am^2 - an^2}{2am - 2an}$
 $= \frac{a(m+n)(m-n)}{2a(m-n)}$

$$= \frac{1}{2}(m+n)$$

Equation of chord MN:

$$y - am^2 = \frac{1}{2}(m+n)(x - 2am)$$

$$2y - 2am^2 = (m+n)x - 2am^2 - 2amn$$

$$2y = (m+n)x - 2amn$$

$$\therefore y = \frac{1}{2}(m+n)x - amn$$

ii. midpoint_{MN} = $\left(\frac{2am + 2an}{2}, \frac{am^2 + an^2}{2} \right)$
 $= \left[a(m+n), \frac{a(m^2 + n^2)}{2} \right]$

iii. $(0, 2)$ satisfies eqn i.
 $2 = 0 - amn$
 $mn = -\frac{2}{a}$

from ii.

$$x = a(m+n) \quad y = \frac{a(m^2 + n^2)}{2}$$

$$\frac{x}{a} = m+n \quad \frac{2y}{a} = m^2 + n^2$$

$$(m+n)^2 = m^2 + 2mn + n^2$$

$$\left(\frac{x}{a} \right)^2 = \frac{2y}{a} + 2x - \frac{2}{a}$$

$$\frac{x^2}{a^2} = \frac{2y}{a} - \frac{4}{a}$$

$$x^2 = 2ay - 4a$$

$$\therefore x^2 = 2a(y - 2)$$

Question 6

a) $P(x) = ax(x+4)(x-4)$

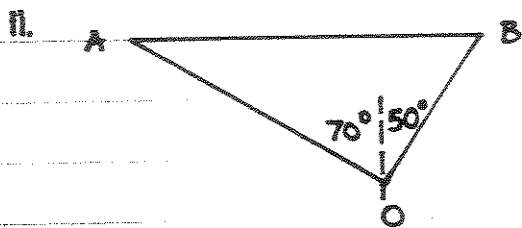
$$P(3) = 21$$

$$3a(3+4)(3-4) = 21$$

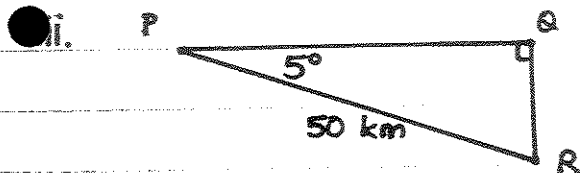
$$a = -1$$

$$\therefore P(x) = -x(x+4)(x-4)$$

b) i. 300 km in 1 h
 $= \frac{300}{6} \text{ km in 10 min}$
 $= 50 \text{ km}$



$$\angle AOB = 70^\circ + 50^\circ = 120^\circ$$

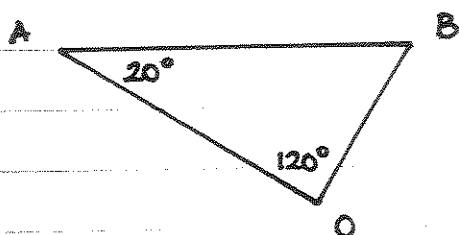


$$\cos 5^\circ = \frac{PQ}{50}$$

$$PQ = 50 \cos 5^\circ$$

but $PQ = AB$

so $AB = 50 \cos 5^\circ$



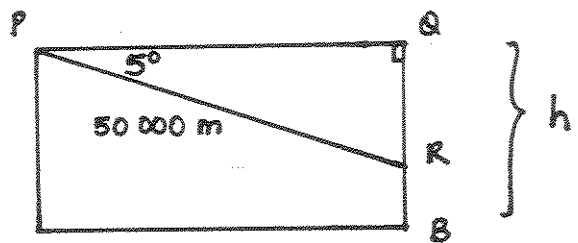
$$\frac{OB}{\sin 20^\circ} = \frac{50 \cos 5^\circ}{\sin 120^\circ}$$

$$OB = \frac{50 \cos 5^\circ \sin 20^\circ}{\sin 120^\circ}$$

$$= 19.6714.. \text{ km}$$

$$= 19671.4.. \text{ m}$$

$$\therefore OB = 19670 \text{ m}$$



$$\sin 5^\circ = \frac{QR}{50\,000}$$

$$QR = 50\,000 \sin 5^\circ$$

$$\tan 2^\circ = \frac{RB}{19670}$$

$$RB = 19670 \tan 2^\circ$$

$$h = QR + RB$$

$$= 5044.67..$$

$$\therefore \underline{\underline{h = 5040 \text{ m}}}$$

c) $f(x) = u(x) - \ln[u(x) + 1]$

i. $f'(x) = u'(x) - \frac{u'(x)}{u(x) + 1}$
 $= \frac{u'(x)u(x) + u'(x) - u'(x)}{u(x) + 1}$
 $= \frac{u'(x)u(x)}{u(x) + 1}$

ii. $\int_0^{\frac{\pi}{2}} \frac{\cos x \sin x}{1 + \sin x} dx$
 i.e. $u(x) = \sin x$

$$\begin{aligned}
 &= \left[\sin x - \ln [\sin x + 1] \right]_0^{\frac{\pi}{2}} \\
 &= \left[\sin \frac{\pi}{2} - \ln \left[\sin \frac{\pi}{2} + 1 \right] \right] \\
 &\quad - \left[\sin 0 - \ln [\sin 0 + 1] \right] \\
 &= \underline{\underline{1 - \ln 2}}
 \end{aligned}$$

Question 7

$$\begin{aligned}
 \text{a) i. } \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 &= \frac{d}{dx} \left(\frac{1}{2} \times \frac{16}{x^2} \right) \\
 &= \frac{d}{dx} (8x^{-2}) \\
 &= -16x^{-3}
 \end{aligned}$$

when $x = 1$

$$\underline{\underline{\ddot{x} = -16 \text{ m/s}^2}}$$

$$\begin{aligned}
 \text{ii. } \frac{dx}{dt} &= \frac{4}{x} \\
 \frac{dt}{dx} &= \frac{x}{4} \\
 t &= \frac{x^2}{8} + c
 \end{aligned}$$

when $t = 0$, $x = 8$

$$0 = \frac{8^2}{8} + c$$

$$c = -8$$

$$t = \frac{x^2}{8} - 8$$

$$8t = x^2 - 64$$

$$x^2 = 8t + 64$$

$$x = \pm \sqrt{8t + 64}$$

$$\therefore \underline{\underline{x = \sqrt{8t + 64}}} \quad \left(\begin{array}{l} \text{given} \\ x > 0 \end{array} \right)$$

iii. when $t = 2$

$$x = \sqrt{8 \cdot 2 + 64}$$

$$= \sqrt{80}$$

$$= \underline{\underline{4\sqrt{5}}}$$

iv. The particle is moving to the right from $x = 8$ and is slowing down (since $v > 0$ and $a < 0$). The particle continues to slow but does not come to rest (since $v \neq 0$).

$$\begin{aligned}
 \text{b) i. } x &= Vt \cos \theta \\
 y &= Vt \sin \theta - \frac{1}{2} g t^2
 \end{aligned}$$

$$\dot{y} = V \sin \theta - gt$$

At max pt: $\dot{y} = 0$

$$0 = V \sin \theta - gt$$

$$t = \frac{V \sin \theta}{g}$$

$$y = V \times \frac{V \sin \theta}{g} \times \sin \theta$$

$$- \frac{1}{2} g \times \left(\frac{V \sin \theta}{g} \right)^2$$

$$= \frac{V^2 \sin^2 \theta}{g} - \frac{V^2 \sin^2 \theta}{2g}$$

$$= \frac{2V^2 \sin^2 \theta}{2g} - \frac{V^2 \sin^2 \theta}{2g}$$

$$h = \frac{V^2 \sin^2 \theta}{2g}$$

ii. At T : $y = \frac{-V^2 \sin^2 \theta}{2g}$

$$\frac{-V^2 \sin^2 \theta}{2g} = Vt \sin \theta - \frac{1}{2}gt^2$$

$$-V^2 \sin^2 \theta = 2g Vt \sin \theta - g^2 t^2$$

$$g^2 t^2 - 2g Vt \sin \theta - V^2 \sin^2 \theta = 0$$

$$t = \frac{2g V \sin \theta \pm \sqrt{4g^2 V^2 \sin^2 \theta + 4g^2 V^2 \sin^2 \theta}}{2g^2}$$

$$= \frac{2g V \sin \theta \pm \sqrt{8g^2 V^2 \sin^2 \theta}}{2g^2}$$

$$= \frac{2g V \sin \theta \pm 2\sqrt{2} g V \sin \theta}{2g^2}$$

$$= \frac{V \sin \theta \pm \sqrt{2} V \sin \theta}{g}$$

$$\therefore t = \frac{V \sin \theta (1 + \sqrt{2})}{g} \quad (t > 0 \text{ only})$$

iii. $x = Vt \cos \theta$

$$= V \times \frac{V \sin \theta (1 + \sqrt{2})}{g} \times \cos \theta$$

$$= \frac{V^2 (1 + \sqrt{2}) \sin \theta \cos \theta}{g}$$

$$\therefore x = \frac{V^2 (1 + \sqrt{2}) \sin 2\theta}{2g}$$